

SAVEETHA ENGINEERING COLLEGE

(Autonomous Institution — Affiliated to Anna University)

SUBJECT: Prompt Engineering

EXPERIMENT NO. 8

Exploration of Prompting Techniques for Video Generation

STUDENT DETAILS

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Aim

To explore and understand various prompting techniques used for generating videos through AI models.

Objective

To demonstrate how different prompt structures, such as simple prompts and detailed prompts, affect the quality, coherence, creativity, motion, and style of AI-generated videos.

AI Video Generation Tools Used

- RunwayML
- Pika Labs
- Synthesia
- Kaiber AI
- Canva AI Video
- Luma AI

Introduction

Artificial Intelligence (AI) video generation tools create videos from text prompts. The quality of generated videos largely depends on how the prompts are written. Simple prompts provide basic instructions, whereas detailed prompts include information about the scene, characters, camera movement, lighting, style, and animation effects.

This experiment investigates how different prompting techniques influence the output generated by AI video creation tools.

Prompt 1: Simple Prompt

Prompt Used

Prompt: *A butterfly flying in a garden.*

Generated Video Description

- Butterfly appears in a garden environment.
- Basic movement and animation.
- Limited background details.
- Minimal camera motion.

Video Link

[Insert AI-generated video link here]

Screenshot

[Insert Screenshot of Generated Scene]

Prompt 2: Detailed Prompt

Prompt Used

Prompt: *Create a cinematic 10-second video of a colorful monarch butterfly flying gracefully through a vibrant flower garden during golden hour. Include blooming roses, sunflowers, and green leaves moving gently in the wind. Use soft sunlight, realistic shadows, depth of field, smooth camera tracking, slow-motion butterfly wing movements, ultra-realistic animation, 4K quality, and natural environmental effects.*

Generated Video Description

- Highly realistic butterfly animation.
- Rich garden details.
- Smooth camera tracking.
- Professional cinematic appearance.
- Better lighting and visual effects.

Video Link

[Insert AI-generated video link here]

Screenshot

[Insert Screenshot of Generated Scene]

Prompt 3: Creative Storytelling Prompt

Prompt Used

Prompt: *A magical butterfly made of glowing blue light flies through an enchanted forest at night. Fireflies illuminate the surroundings while sparkling particles follow the butterfly. The camera slowly follows the butterfly as it passes ancient trees and glowing flowers. Fantasy style, cinematic atmosphere, smooth transitions, ultra-HD quality.*

Generated Video Description

- Fantasy-themed environment.
- Creative visual effects.
- Dynamic motion and transitions.
- Enhanced storytelling elements.

Video Link

[Insert AI-generated video link here]

Screenshot

[Insert Screenshot of Generated Scene]

Comparison Between Simple and Detailed Prompts

Feature	Simple Prompt	Detailed Prompt
Video Quality	Basic	High Quality
Scene Details	Limited	Rich and Detailed
Camera Motion	Minimal	Smooth Cinematic Motion
Lighting Effects	Generic	Realistic Lighting
Animation Quality	Basic	Advanced Animation
Creativity	Moderate	High
Scene Consistency	Medium	High
Realism	Low	High
User Control	Limited	Extensive

Observations

- Simple prompts generated videos quickly but provided limited control over visual details.
- Detailed prompts produced significantly better video quality and more accurate scene representation.
- Including camera movement instructions improved cinematic effects.
- Lighting and environmental descriptions enhanced realism.
- Creative storytelling prompts resulted in unique and visually appealing videos.
- Detailed prompts increased scene consistency and reduced unexpected visual elements.
- AI models responded better when prompts specified style, quality, duration, motion, and atmosphere.

Analysis

The experiment demonstrates that prompt engineering plays a crucial role in AI video generation. Simple prompts provide general results, while detailed prompts allow precise control over:

- Visual appearance
- Character movement
- Camera angles
- Lighting conditions
- Animation style
- Storytelling elements

As the level of prompt detail increased, the generated videos showed improvements in realism, coherence, creativity, and overall quality.

Advantages of Detailed Prompting

- Better scene consistency
- Improved visual quality
- More realistic animations
- Greater creative control
- Enhanced storytelling
- Professional-looking output

Limitations

- Longer prompts may require more generation time.
- Results can vary across different AI video tools.
- Some AI models may interpret prompts differently.
- High-quality generation may require premium subscriptions.

Result

Thus, the prompting techniques for AI video generation were successfully explored and analyzed using multiple prompt structures. The experiment demonstrated that detailed prompts significantly improve video quality, scene consistency, animation style, motion, transitions, and creativity compared to simple prompts.